

Discrete Mathematics

Graph Theory



Incidence Matrix

Let $G = (V, E)$ be a graph with $v_1, v_2, \dots, v_n$ vertices and $e_1, e_2, \dots, e_m$ edges. Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix

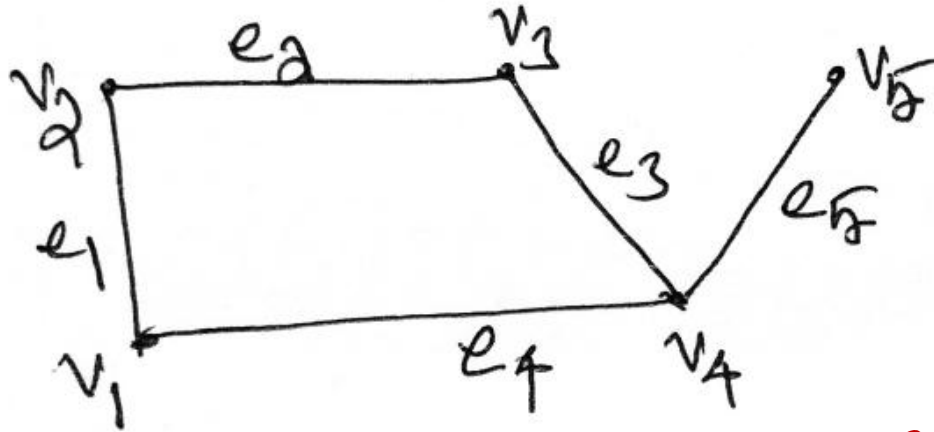
$$M = [m_{ij}]$$

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i \\ 0 & \text{otherwise} \end{cases}$$

Note

The number of 1's in each row equals to the degree of the corresponding vertex.

Find the incidence matrix of the following graph.



Soln

Incidence matrix =

$$\begin{array}{c}
 \begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array}
 \begin{array}{c} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{array}
 \begin{bmatrix}
 1 & 0 & 0 & 1 & 0 \\
 1 & 1 & 0 & 0 & 0 \\
 0 & 1 & 1 & 0 & 0 \\
 0 & 0 & 1 & 1 & 1 \\
 0 & 0 & 0 & 0 & 1
 \end{bmatrix}
 \end{array}$$

Connectivity

Walk

A walk in a graph G_1 is defined as an alternating sequence of vertices and edges

$v_0, e_1, v_1, e_2, v_2, \dots, e_n, v_n$

beginning and ending with vertices such that each line e_i is incident with v_{i-1} and v_i .

A walk joining v_0 and v_n is called v_0-v_n walk.

Here, v_0 is called the *initial vertex* and v_n is called the *terminal vertex* of the walk.

- The number of edges in the walk is known as the length of the walk.
- The walk is also denoted by $v_0v_1\dots v_n$.

Trail

A walk is called a trail if all its edges are distinct.

Path

A walk is called a path if all its vertices are distinct.

Note, circuit

A circuit is defined as a closed path of non-zero length that does not contain a repeated edge.

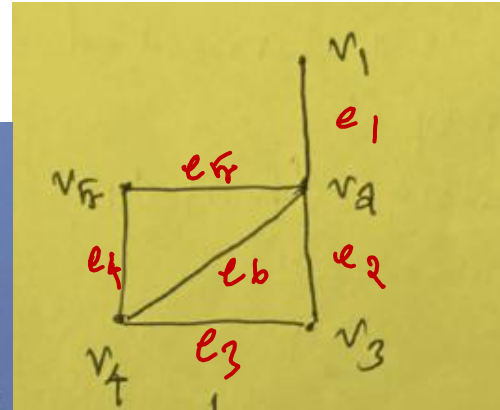
In the above graph

walk: v_1, v_2, v_3, v_4, v_5 is a walk

Trail: $v_1, v_2, v_4, v_3, v_2, v_5$ [here all edge distinct]

path: v_1, v_2, v_4, v_5 [here all vertex distinct]

circuit: v_2, v_3, v_4, v_5, v_2 [closed path]



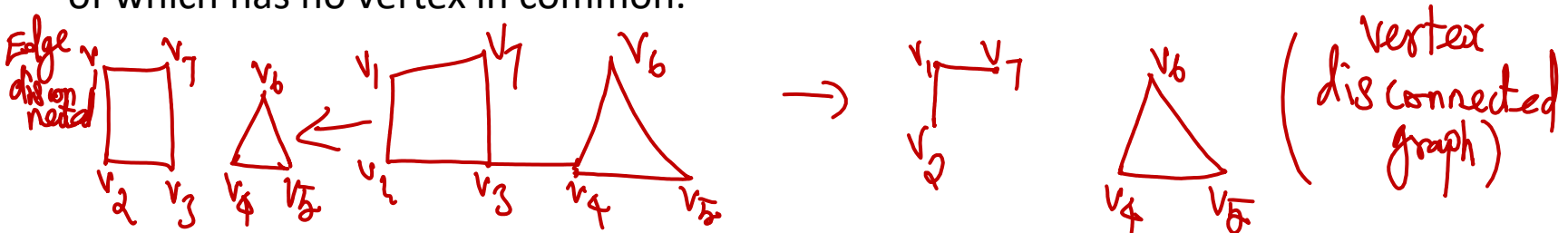
Connectedness and Components: A graph is said to be **connected** if it is possible to travel from any vertex to any other vertex along a sequence of adjacent edges of the graph.



Connected Vertices: Two vertices u and v of a graph G_1 are said to be connected if there exists a path from u to v .

Connected Graph: A graph G_1 is said to be connected if every pair of vertices in G_1 are connected.

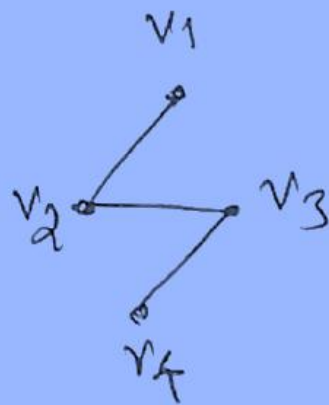
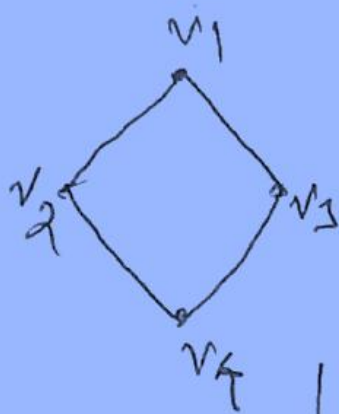
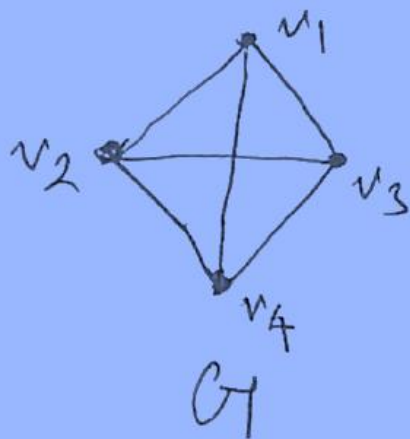
Disconnected Graph: A graph which is not connected is said to be **disconnected**. A disconnected graph is the union of two or more connected subgraphs, each pair of which has no vertex in common.



Note

A graph H is a Connected Component of G if and only if

- H is a subgraph of G
- H is connected



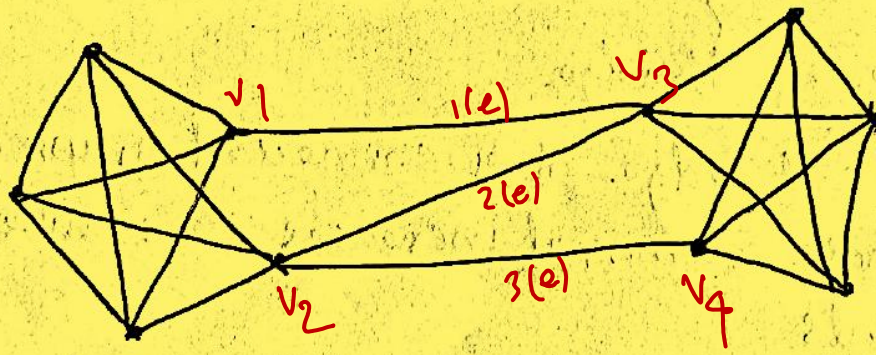
(Connected Components of G)

Edge connectivity

Let G be a connected graph. The edge connectivity of G is defined as the minimum number of edges whose removal results in a disconnected graph. The edge connectivity of a connected graph G is denoted by $\lambda(G)$.

vertex connectivity

Let G be a connected graph. The vertex connectivity of G is defined as the minimum number of vertices whose removal results in a disconnected graph. The vertex connectivity of a connected graph G is denoted by $\kappa(G)$.



here edge connectivity $\lambda(G) = 3$
 vertex connectivity $\kappa(G) = 2$

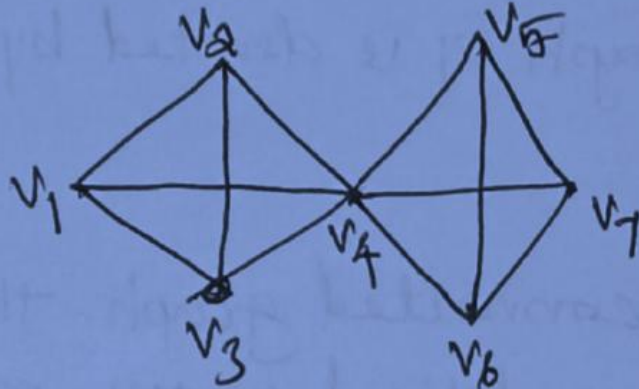
note

The vertex connectivity of a graph G is less than or equal to the edge connectivity of G

ie $\kappa(G) \leq \lambda(G)$

Distance

Let u and v be any two vertices of the connected graph G . The distance between u and v is denoted by $d(u, v)$ and is defined as the length of the shortest path.



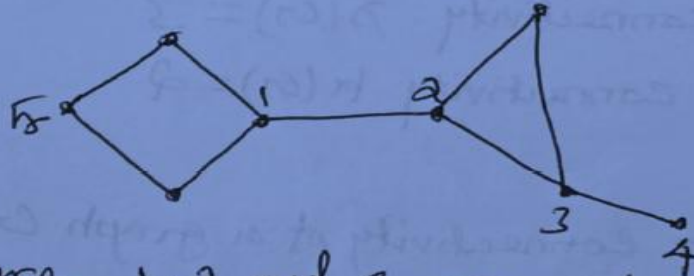
$$d(v_1, v_6) = 2 \quad [v_1 - v_4 - v_6 \text{ shortest path}]$$

Cut vertex

A cut vertex of a connected graph G is a vertex whose removal increases the number of components

Cut edge or Bridge

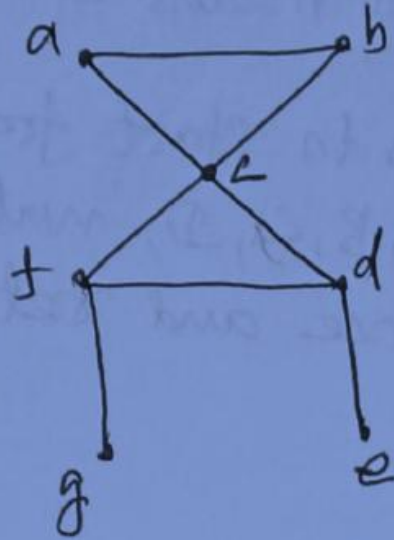
A cut edge of a connected graph G is an edge whose removal increases the number of components



here 1, 2 and 3 are endpoints
the lines $\{1, 2\}$ and $\{3, 4\}$ are bridges.
5 is non-endpoint

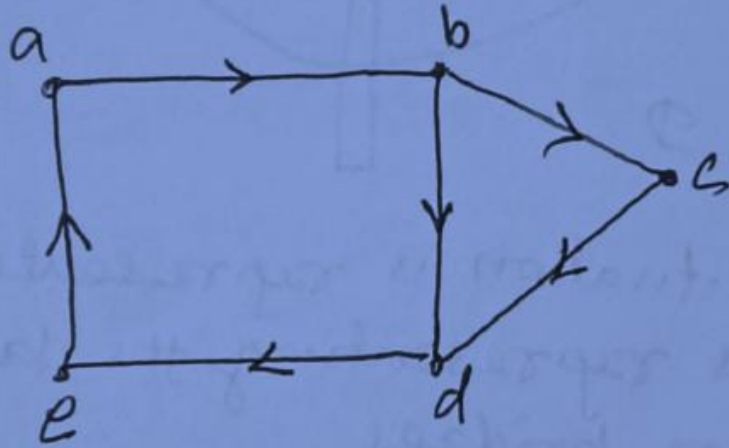
Note

1. An undirected graph is called connected if there is a path between every pair of ~~x~~ distinct vertices of the graph.

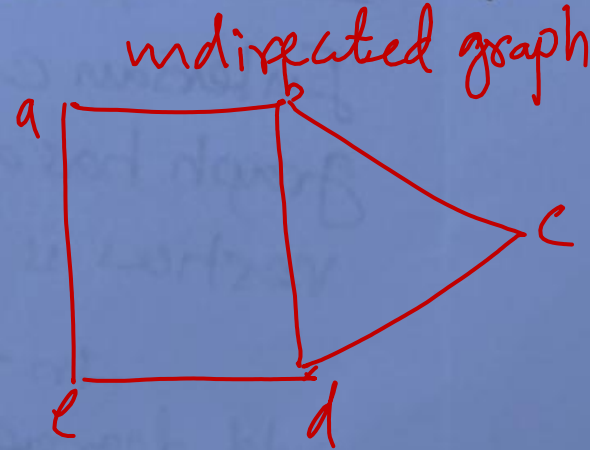
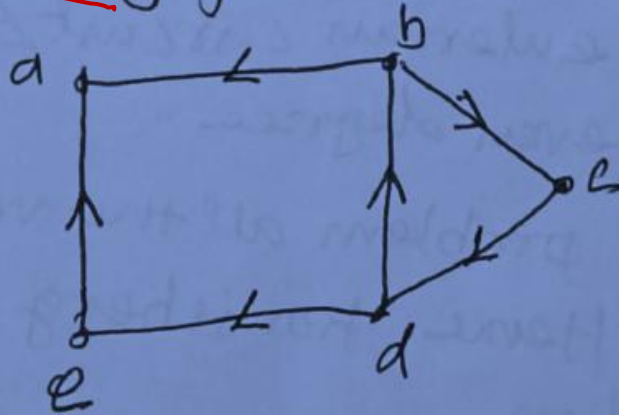


path a to e
1. a-c-d-e
2. a-c-f-d-e
3. a-b-c-d-e

2. A directed graph is strongly connected if there is a path from a to b and from b to a whenever a and b are vertices in the graph



3. A directed graph is weakly connected if there is a path between every two vertices in the underlying undirected graph.



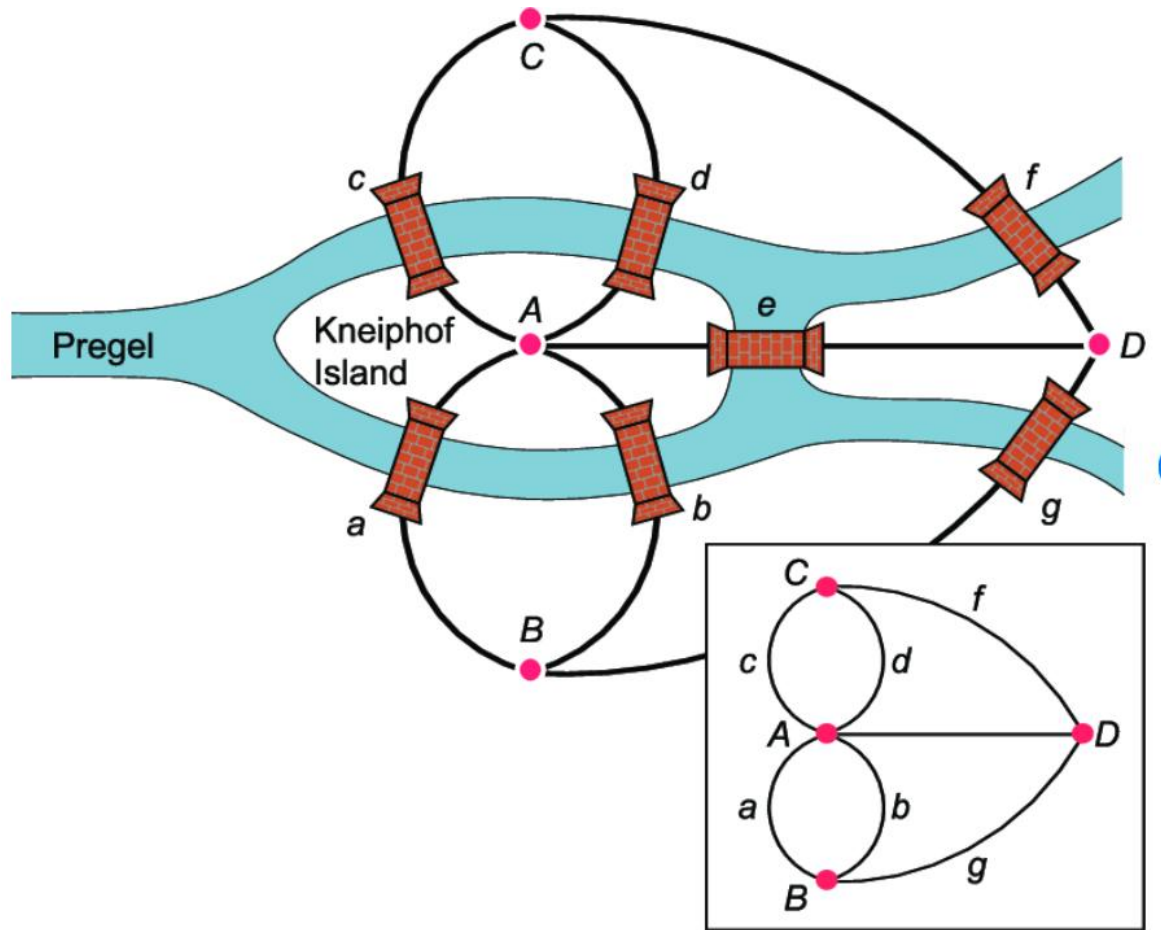
Konigsberg Bridge Problem

- There are two islands **A** and **D** formed by a river.
They are connected to each other and to the river banks **C** and **B** by means of 7 bridges.
- The problem is to start from any one of the 4 land areas **A, B, C, D**, walk across each bridge exactly once and return to the starting point.
- When the situation is represented by a graph,
with vertices representing the land areas and the edges as bridges.

Note:

The problem is to find whether there is an Eulerian circuit in the graph.
But a connected graph has an Eulerian circuit if and only if each of its vertices is of even degree.

- In the problem, all the vertices are of **odd degree**.
Hence, the **Königsberg Bridge Problem** has **no solution**.



Eulerian Graph

A trail in G is called an Eulerian trail if it contains all the edges of G exactly once and it contains all the vertices of G .

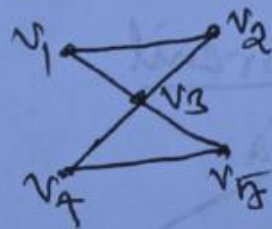
A closed Eulerian trail is called Eulerian circuit.

A graph having an Eulerian circuit is called an Eulerian graph.

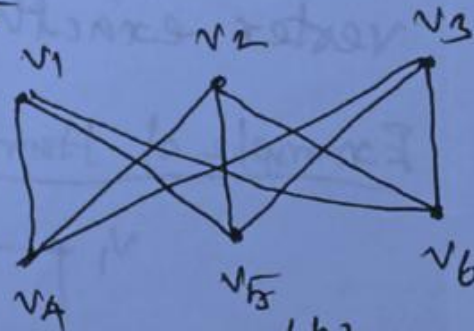
Note

1. A connected graph G is Eulerian if and only if its vertices are all of even degree.
2. A path in a graph G is called an Euler path if it includes every edge exactly once. Hence Euler path is also called Euler trail.
3. A closed Euler path is called Euler circuit.

1. Let G be the following graph. verify whether G has an Eulerian circuit or not



(a)



(b)

$K_{3,3}$

Soln:

Given graph (a) is connected and all the vertices are of even degree

$$\deg(v_1) = \deg(v_2) = \deg(v_3) = \deg(v_4) = \deg(v_5) = 2$$

Thus G has Eulerian circuit

$$v_1 - v_3 - v_5 - v_4 - \underline{v_3} - v_2 - v_1 \quad (\text{edge different})$$

In graph (b) all the vertices are of degree 3. Hence, it does not contain an Eulerian circuit.

Hamiltonian graph

A circuit in a graph G is called a hamiltonian circuit if it contains each vertex in G exactly once, except for the starting and ending vertex that appears twice.

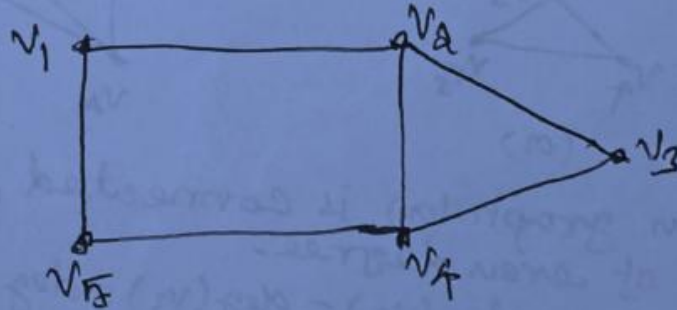
A graph G is called a Hamiltonian graph if it contains a hamiltonian circuit

A Hamiltonian path is a path that contains all vertices of G where the end points may be distinct.

Note

An Eulerian circuit traverses every edge exactly once, but vertices may repeat. on the other hand, Hamiltonian circuit visits each vertex exactly once, but edges may repeat

Example of Hamiltonian Circuit

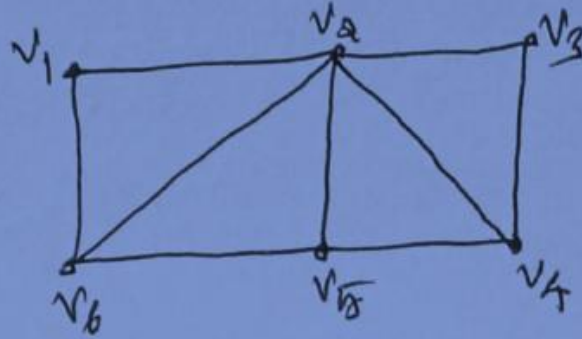


The above graph has a hamiltonian circuit namely $v_1 - v_2 - v_3 - v_4 - v_5 - v_1$

2 Give an example of a graph which is Hamiltonian but not Eulerian vice versa

Soln

Hamiltonian but not Eulerian



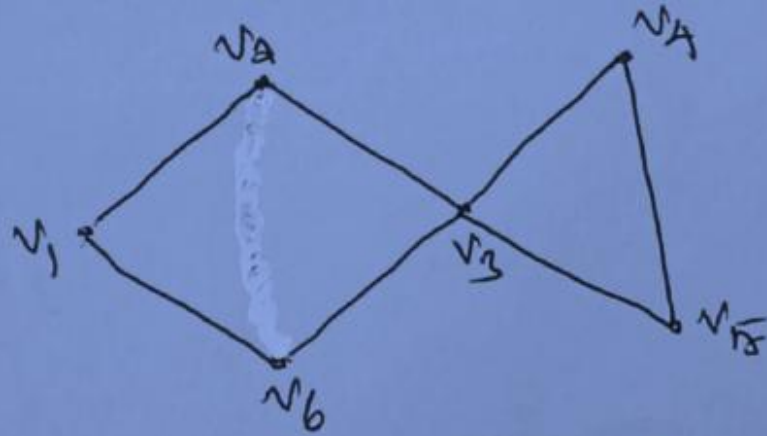
The graph contains a Hamiltonian circuit

$$v_1 - v_2 - v_3 - v_4 - v_5 - v_6 - v_1$$

Since degree of each vertex is not even

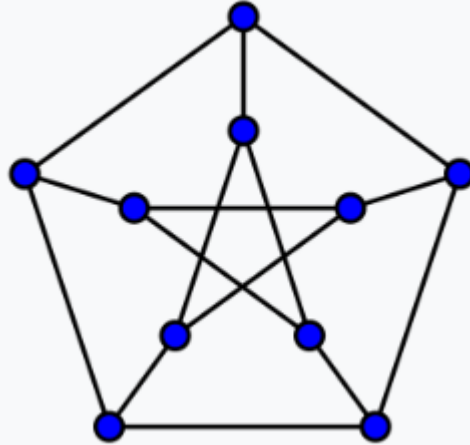
$\therefore$ the above graph is not Eulerian

Eulerian but not Hamiltonian



Since the degree of each vertex is even
the above graph is Eulerian
but, it does not contain a Hamiltonian
Circuit.

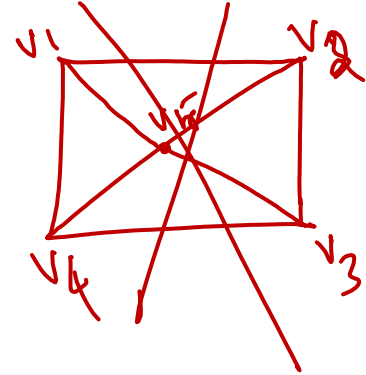
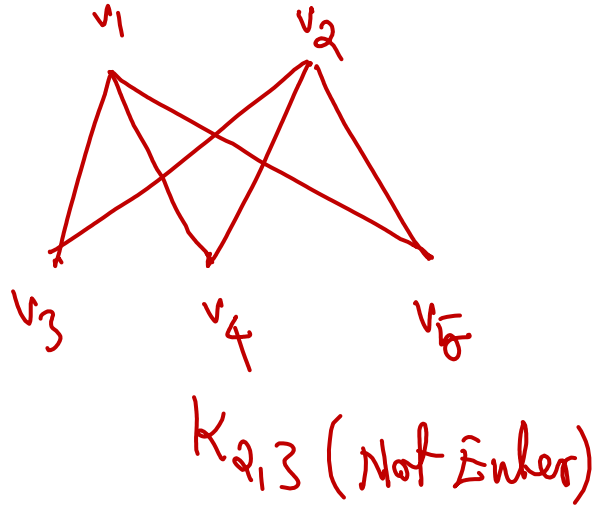
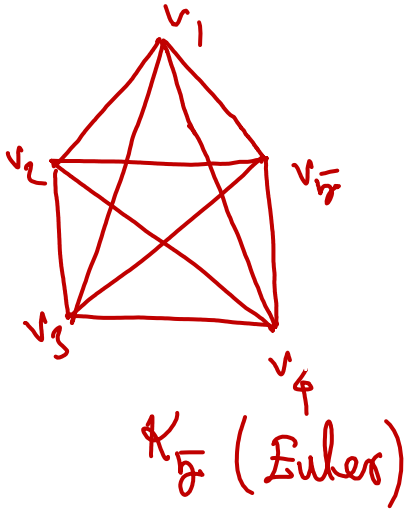
Petersen graph



The **Petersen graph** is an undirected graph with 10 vertices and 15 edges.

Which of the following graphs are Eulerian

1. Complete graph K_5
2. Complete bipartite graph $K_{2,3}$
3. Peterson graph. (Not Euler)



The maximum number of edges in a bipartite graph on 12 vertices is 36.

Soln $\rightarrow$ bipartite graph

$$\text{Int}(V) = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8, v_9, v_{10}, v_{11}, v_{12}\}$$
$$V_1 = \{v_1, v_3, v_5, v_7, v_9, v_{11}\}$$
$$V_2 = \{v_2, v_4, v_6, v_8, v_{10}, v_{12}\}$$

$\rightarrow$ partition

$$\text{Total no of edges} = 6 \times 6 = 36$$

